

# Reflection: What Mathematics Contributes

LING487 course synthesis

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# The course began with one question

## Question

What does mathematics contribute to linguistic explanation—not merely to calculation after an analysis?

- It makes abstractions explicit.
- It identifies invariants and admitted variants.
- It exposes consequences and hidden domain assumptions.
- It distinguishes possible structures from empirically sustained ones.

# Where we are

1. Connections
2. Linguistic reality
3. Reflection

# The same idea appeared under several names

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|                    |                                                       |
|--------------------|-------------------------------------------------------|
| <b>Membership</b>  | an object satisfies a set condition                   |
| <b>Application</b> | an argument satisfies a functional dependency         |
| <b>Composition</b> | domains connect across functions                      |
| <b>Derivation</b>  | a rule licenses the next formal object                |
| <b>Acceptance</b>  | a computation reaches an admitted final configuration |
| <b>Validation</b>  | a model connects adequately to linguistic evidence    |

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## Take-away

Admissibility is the thread from basic notation to explanation.

# Description and method must not be collapsed

| String family           | Description              | Method                  |
|-------------------------|--------------------------|-------------------------|
| <b>Regular</b>          | regexp / regular grammar | finite automaton        |
| <b>Context-free</b>     | context-free grammar     | PDA / CKY               |
| <b>Regular relation</b> | paired regexp            | finite-state transducer |

## Question

What becomes confused when a grammar is called a parser?

# Formal equivalence can conceal explanatory difference

- Two grammars can generate the same stringset and assign different structures.
- An NFA and DFA can accept the same strings and retain different state interpretations.
- Equivalent regexps can expose different generalizations.
- Binarization can preserve recognition while adding method-specific nodes.

## Take-away

Always state what notion of sameness the equivalence preserves.

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# Three fundamental questions remain empirical constraints

- ① Ontogeny: how can children acquire a usable system?
- ② Continuity: how are child and adult grammars related?
- ③ Diversity: why do languages vary so extensively within limited dependency types?

## Question

Which formal result from the course bears on each question, and what additional evidence is required?



# Grammar is indispensable without being sufficient

- Grammatical structure constrains possible interpretation and action.
- Context, knowledge, social relation, attention, and material conditions also constrain success.
- Grammar can therefore be an INUS component of linguistic activity.

## Guided analysis

Construct one sufficient package for a successful request. Replace one component while preserving success.

# Contingent homomorphisms preserve what languages actually sustain

$$h(x \star y) = h(x) \circ h(y)$$

is a mathematical possibility. Its linguistic force depends on:

- how  $x, y, \star, h$ , and  $\circ$  are interpreted;
- whether the combinations are grammatically admitted;
- whether speakers acquire and use the correspondence;
- whether comparison across languages preserves the proposed invariant.

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# Five questions to ask of any formal linguistic analysis

- ① What objects and relations have been abstracted?
- ② What is invariant, and what variation is admitted?
- ③ Is this a description, computation, model, or explanation?
- ④ Which domains must connect for the operations to be defined?
- ⑤ What evidence validates the correspondence to linguistic reality?

# Final reflection

Choose one analysis you encountered before this course.

- ① Restate it using the course vocabulary.
- ② Identify one hidden assumption.
- ③ Distinguish its descriptive and explanatory claims.
- ④ Say which variant would test its proposed invariant.
- ⑤ State one result that would make you revise it.

# The destination is better linguistic questions

*Mathematics earns its place in linguistics when it clarifies what we are claiming about language, what follows from the claim, and how reality can answer back.*

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